gpeyt1-45 · Elastic settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A flexible 9 ft × 9 ft mat for an equipment pad applies Δp = 3,200 psf to a deep, homogeneous dense gravelly sand with Em = 420,000 psf (≈ 2,917 psi) and ν = 0.3. Estimate the AVERAGE immediate settlement under the flexible pad (not the center).

Most nearly, the average elastic settlement is:

A. 0.71 in.
B. 0.84 in.
C. 0.78 in.
D. 0.06 in.

Given equations / data

> δv = Cd Δp Bf (1 − ν²)/Em; Cd (square, flexible): center 1.12, corner 0.56, average 0.95; square rigid 0.99

Worked solution

1. 1 − ν² = 0.91
2. δv = 0.95 × 3,200 × 9 × 0.91/420,000 = 0.0593 ft = 0.71 in.

Final answer

0.71 in.

Why the other values are wrong

- **0.84 in.** — used the center factor 1.12
- **0.78 in.** — omitted (1 − ν²)
- **0.06 in.** — left the answer in ft labeled in.

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.2 Settlement (Elastic Method) and Cd table, p. 95

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.